SENSOR CALIBRATION AND QUANTISATION

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Convert raw analogue-to-digital converter (ADC) counts into engineering units by calibrating a slide potentiometer against a ruler.	☐
2	Calculate the resolution of a measurement from the ADC resolution and the sensor calibration, and compare it with the calibration error.	☐
3	Calibrate the ultrasonic distance sensor, and demonstrate aliasing when a signal is sampled too slowly.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware set-up described in the practical exercises, Slide Potentiometer and Ultrasonic Sensor, which detail how each sensor is connected to the Arduino Uno and read into Simulink. The set-up is not repeated here.

A ruler marked in millimetres and a flat piece of card, at least 10 cm square, are also needed. Fix the ruler alongside the slide potentiometer with tape so that the position of the slider can be read accurately.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. Each sensor is read at several known positions, and a calibration line is fitted in MATLAB to convert the raw readings into millimetres or centimetres. The resolution and accuracy of each measurement are then compared, and the effect of sampling too slowly is investigated. The exercise supports the ACE-CORE learning guide, Microcontroller-Based Measurement for Control Systems.

Background: Quantisation and Calibration

The ADC of the Arduino Uno converts a voltage between 0 V and 5 V into an integer count *N* between 0 and 1023. The smallest change in voltage it can resolve, its resolution, is:

$$\Delta V = \frac{5\ \mathrm{V}}{1024} \approx 4.9\ \mathrm{mV\ per\ count}$$

A sensor that is close to linear can be calibrated with a straight line, where *x* is the physical quantity, *m* is the slope and *c* is the offset. The line is fitted to measured pairs of (*N*, *x*) by least squares. The slope gives the resolution in engineering units, in mm per count for the slide potentiometer:

$$x = m\,N + c$$

Calibrate the Slide Potentiometer

1. Open the model from the practical exercise, Slide Potentiometer, and log the raw ADC count (before any scaling) to the workspace.
2. Move the slider to the first marked position, wait one second, and record the position from the ruler and the ADC count in Table 2. Average a few readings if the count changes slightly.
3. Repeat for at least six positions spread over the full travel of the slider.
4. Enter the readings into the script below and run it to fit the calibration line.

Table 2: *Calibration Readings for the Slide Potentiometer*

Point	Position from ruler (mm)	ADC count, *N*	Position from fit (mm)	Error (mm)
1				
2				
3				
4				
5				
6				

MATLAB Script: fitting a calibration line to the slide potentiometer readings

```
%% Enter your readings from the calibration table
x_mm = [ ];          % positions from the ruler [mm]
N    = [ ];          % ADC counts at each position

p = polyfit(N, x_mm, 1);            % straight-line fit: x = m*N + c
m = p(1), c = p(2)                  % slope [mm/count] and offset [mm]
x_fit = polyval(p, N);
err = x_mm - x_fit;                 % calibration error [mm]
fprintf('Resolution = %.3f mm per count\n', abs(m));
fprintf('Largest calibration error = %.2f mm\n', max(abs(err)));

plot(N, x_mm, 'ko', N, x_fit, 'b-'); grid on;
xlabel('ADC count, N'); ylabel('Position [mm]');
legend('Measured', 'Calibration line', 'Location', 'southeast');
```

Compare the resolution with the largest calibration error. If the error is much larger than the resolution, the accuracy of the measurement is limited by the sensor and the calibration, not by the ADC.

Calibrate the Ultrasonic Sensor

1. Open the model from the practical exercise, Ultrasonic Sensor, and log the measured distance to the workspace.
2. Hold the card square to the sensor at five distances between 10 cm and 50 cm, measured from the front of the sensor with the ruler. Record the true and measured distances in Table 3.
3. Fit a correction line of true distance against measured distance with `polyfit`, as for the slide potentiometer.
4. Repeat one reading with the card tilted by about 30°, and one with a soft target such as a cloth. Note what happens to the measurement.

Table 3: *Calibration Readings for the Ultrasonic Sensor*

True distance (cm)	Measured distance (cm)	Error (cm)	Corrected distance (cm)
10			
20			
30			
40			
50			

Observe Aliasing

A signal must be sampled more than twice per cycle of its fastest component to be captured correctly, which is known as the Nyquist criterion. When it is sampled more slowly, the samples can describe a much slower signal that is not really

there, which is called aliasing. Figure 1 shows a slider moved back and forth at 1.8 Hz and sampled at 2 Hz: the samples suggest a slow drift at 0.2 Hz.

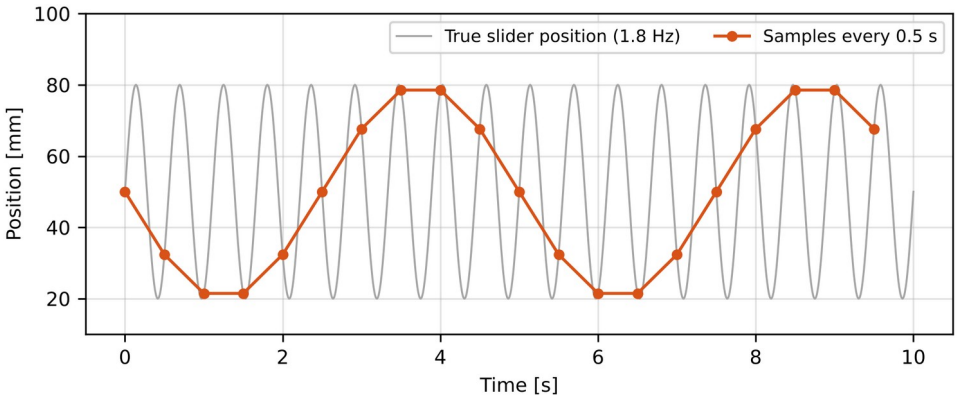


Figure 1: *Theoretical Example of Aliasing: a 1.8 Hz Motion Sampled at 2 Hz*

1. Set the sample time of the slide potentiometer model to 0.5 s and log the position.
2. Move the slider back and forth over most of its travel, about twice per second, for 10 s. Compare the logged signal with what you did.
3. Repeat with a sample time of 0.02 s, and explain the difference between the two recordings.

4.0 Concluding Remarks

This practical exercise calibrated two sensors read by the Arduino Uno. Using MATLAB, Simulink and ACE-Lab hardware, raw readings were converted into engineering units with a least-squares calibration line, and the resolution set by the 10-bit ADC was compared with the error of the calibration.

The exercise also showed that a measurement is only as good as its sampling: a signal sampled fewer than two times per cycle can appear as a slower signal that does not exist. In a control system, the sample time must therefore be short enough to capture the fastest changes that matter in the measured signal.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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